

Fourier evaluation proves spectral inclusion for free complex Banach lattices

Solution packet for arXiv:2207.08090

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Outcome. For every complex Banach space E and every bounded operator $T : E \rightarrow E$, the induced lattice homomorphism $\bar{T} : \text{FBL}_{\mathbb{C}}[E] \rightarrow \text{FBL}_{\mathbb{C}}[E]$ satisfies

$$\sigma(T) \subseteq \sigma(\bar{T}).$$

The missing complex phase is recovered by the first Fourier coefficient of the circle of evaluation functionals associated with an eigenfunctional of T^* .

Source question

The source constructs $\text{FBL}_{\mathbb{C}}[E]$ as a complexification of a real free Banach lattice. Its canonical embedding $\delta_E : E \rightarrow \text{FBL}_{\mathbb{C}}[E]$ obeys

$$\delta_E(x)(\text{Re } z^*) = z^*(x), \quad x \in E, \quad z^* \in E^*.$$

For $T : E \rightarrow E$, the universal property produces $\bar{T}$ with $\bar{T}\delta_E = \delta_E T$. In the functional model,

$$(\bar{T}f)(\text{Re } w^*) = f(\text{Re } T^*w^*). \quad (1)$$

The source proves inclusion of approximate point spectra, handles positive residual spectral values, and asks whether the full spectrum is always included. It obtains only $|\lambda| \in \sigma(\bar{T})$ for a general $\lambda \in \sigma(T)$.

□

Remark 6.2. The following facts, which are known in the real case (see [28, Section 3]), can be easily extended to the complex setting:

- (1) T is injective if and only if $\bar{T}$ is injective.
- (2) T has dense range if and only if $\bar{T}$ has dense range.

Taking this into account, the second part of the former proposition can be refined as follows: $0 \in \sigma_p(T)$ if and only if $0 \in \sigma_p(\bar{T})$; $0 \in \sigma_c(T)$ if and only if $0 \in \sigma_c(\bar{T})$; $0 \in \sigma_r(T)$ if and only if $0 \in \sigma_r(\bar{T})$.

Remark 6.3. In general, we can have $\sigma_p(T) \neq \sigma_p(\bar{T})$, $\sigma_a(T) \neq \sigma_a(\bar{T})$ or $\sigma(T) \neq \sigma(\bar{T})$. Consider the operator $T : \mathbb{C} \rightarrow \mathbb{C}$ defined by $Tz = -z$, which has $\sigma(T) = \sigma_a(T) = \sigma_p(T) = \{-1\}$. However, the function $(f + ig) : \mathbb{R}^2 \rightarrow \mathbb{C}$ defined by $(f + ig)(x, y) = \sqrt{x^2 + y^2}$ is a fixed point of $\bar{T}$ and then $1 \in \sigma_p(\bar{T})$.

In view of the results exposed in Proposition 6.1 one may ask whether it is always true that $\sigma(T)$ is contained in $\sigma(\bar{T})$. The remainder of this section will be dedicated to exploring this question. Let us begin by showing that the positive elements of $\sigma(T)$ belong also to $\sigma(\bar{T})$.

Proposition 6.4. *If $\lambda \in \sigma(T) \cap [0, +\infty)$, then $\lambda \in \sigma(\bar{T})$.*

Proof. Since $\sigma(T) = \sigma_a(T) \cup \sigma_r(T)$, by the first part of Proposition 6.1 it is enough to show that $\sigma_r(T) \cap [0, +\infty) \subset \sigma(\bar{T})$. Fix any $\lambda \in \sigma_r(T) \cap [0, +\infty)$. Then $\lambda \in \sigma_p(T^*)$ (see [1, Theorem 6.19]). Take $z^* \in E^*$, $z^* \neq 0$, such that $T^*z^* = \lambda z^*$. For every $f \in \text{FBL}_{\mathbb{C}}[E]$, by the positive homogeneity of f , we have that

$$(\lambda - \bar{T})f(\Re z^*) = \lambda f(\Re z^*) - f(\Re T^*z^*) = \lambda f(\Re z^*) - f(\lambda \Re z^*) = 0,$$

so that $\lambda - \bar{T}$ cannot have dense range and, in particular, $\lambda \in \sigma(\bar{T})$. □

Given $\theta \in \mathbb{R}$, let us denote by M_θ the operator defined on E by $M_\theta z = e^{i\theta} z$, for every $z \in E$. These operators will play an important role in the proof of Proposition

The Fourier-evaluation functional

Lemma 1. *Suppose $z^* \in E^* \setminus \{0\}$ and $T^*z^* = \lambda z^*$. Then λ is an eigenvalue of $\overline{T}^*$.*

Proof. Write $\lambda = re^{i\theta}$ with $r \geq 0$; when $r = 0$, choose $\theta = 0$. For $t \in \mathbb{R}$, let

$$\varepsilon_t(f) = f(\operatorname{Re}(e^{it}z^*)), \quad f \in \operatorname{FBL}_{\mathbb{C}}[E].$$

These evaluations are uniformly bounded: $\|\varepsilon_t\| \leq \|z^*\|$. Moreover $t \mapsto \varepsilon_t(f)$ is continuous, by the functional representation of $\operatorname{FBL}_{\mathbb{C}}[E]$. Hence

$$\Phi(f) = \frac{1}{2\pi} \int_0^{2\pi} e^{-it} f(\operatorname{Re}(e^{it}z^*)) dt \quad (2)$$

defines a bounded linear functional.

It is nonzero. Indeed, the canonical-evaluation identity gives, for every $x \in E$,

$$\Phi(\delta_E x) = \frac{1}{2\pi} \int_0^{2\pi} e^{-it} (e^{it}z^*)(x) dt = z^*(x).$$

Choose x with $z^*(x) \neq 0$.

Using (1), complex linearity of T^* , and positive homogeneity of every free-lattice function under the real scalar r , we obtain

$$\begin{aligned} \Phi(\overline{T}f) &= \frac{1}{2\pi} \int_0^{2\pi} e^{-it} f(\operatorname{Re} T^*(e^{it}z^*)) dt \\ &= \frac{r}{2\pi} \int_0^{2\pi} e^{-it} f(\operatorname{Re}(e^{i(t+\theta)}z^*)) dt \\ &= re^{i\theta} \Phi(f) = \lambda \Phi(f). \end{aligned}$$

The last substitution is $s = t + \theta$, using periodicity. Thus $\overline{T}^*\Phi = \lambda\Phi$ with $\Phi \neq 0$. □

Full spectral inclusion

Theorem 1. *For every complex Banach space E and every bounded operator $T : E \rightarrow E$,*

$$\sigma(T) \subseteq \sigma(\overline{T}).$$

Proof. Fix $\lambda \in \sigma(T)$. If $\lambda \in \sigma_a(T)$, take unit vectors x_k with $(\lambda - T)x_k \rightarrow 0$. Since δ_E is an isometry and intertwines T with $\overline{T}$,

$$\|(\lambda - \overline{T})\delta_E x_k\| = \|\delta_E((\lambda - T)x_k)\| \longrightarrow 0.$$

Thus $\lambda \in \sigma_a(\overline{T}) \subseteq \sigma(\overline{T})$.

Now suppose $\lambda \notin \sigma_a(T)$. Then $\lambda - T$ is bounded below, so its range is closed. It is not onto because $\lambda \in \sigma(T)$; hence its range is a proper closed subspace. Hahn–Banach supplies $z^* \in E^* \setminus \{0\}$ annihilating $(\lambda - T)E$. Equivalently, $T^*z^* = \lambda z^*$. Lemma 1 gives $\lambda \in \sigma_p(\overline{T}^*)$, which implies $\lambda \in \sigma(\overline{T})$. □

Why the averaging is the missing step

A single evaluation at $\operatorname{Re} z^*$ detects $T^*z^* = \lambda z^*$ only when $\lambda \geq 0$, because free-lattice functions are merely positively homogeneous on the real dual. Rotating z^* around its circle and taking the first Fourier coefficient projects precisely onto the complex-linear component: on $\delta_E(E)$ it equals z^* , while T^* rotates the circle by $\arg \lambda$ and dilates it by $|\lambda|$. This restores the full multiplier λ , rather than only its modulus.

Audit notes

- The integral in (2) is a scalar integral of uniformly bounded continuous functions; it defines a bounded complex-linear functional.
- The factor e^{-it} , not e^{it} , is essential: after shifting t by θ it produces $e^{i\theta}$.
- An eigenvalue of $\overline{T}^*$ is sufficient for membership in $\sigma(\overline{T})$; no claim about the point spectrum of $\overline{T}$ is made.
- The argument covers $\lambda = 0$ as well. Then $T^*z^* = 0$ and the Fourier functional is annihilated by $\overline{T}^*$.
- This packet does not address the separate real-scalar question whether nonisomorphic E, F can have lattice-isomorphic $\text{FBL}[E], \text{FBL}[F]$.

Source

D. de Hevia and P. Tradacete, *Free complex Banach lattices*, arXiv:2207.08090 (2022).